Unit 6 Pre-Test

Functions & Linear Relationships

Standards assessed: NC.8.F.1–F.5 · NC.8.EE.8

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.F.1 · NC.8.F.3 · NC.8.F.5 Identify functions and read slope and y-intercept from an equation.

• *I can decide whether a relation is a function and read slope and y-intercept from y = mx + b.*

Match each vocabulary term to its definition, then answer each question.

B function ✓	**A.** Where a graph crosses the y-axis (the value of y when x = 0).
D slope ✓	**B.** A relation in which each input has exactly one output.
A y-intercept ✓	**C.** A point written (x, y) that gives a location on the graph.
C ordered pair ✓	**D.** The steepness of a line, found as rise over run.
E linear function ✓	**E.** A function whose graph is a straight line.
F input ✓	**F.** The x-value that is put into a function.

Slope-intercept form — reference

$$y = mx + b$$

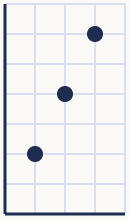
m = *slope (rise over run)* · **b** = *y-intercept (where x = 0)*

1. *Function?*

Is this a function? (1, 2), (2, 4), (3, 6)

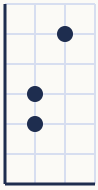
2. *Function?*

Is this a function? (1, 2), (1, 3), (2, 5)



Answer:

answer: yes

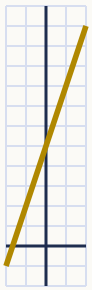


Answer:

answer: no

3. *Slope*

In y = 3x + 5, what is the slope?

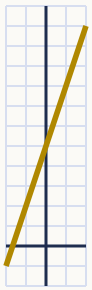


m =

answer: 3

4. *y-intercept*

In y = 3x + 5, what is the y-intercept?

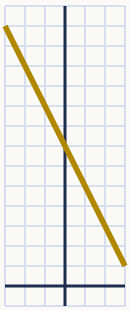


b =

answer: 5

5. *Slope*

In y = −2x + 7, what is the slope?

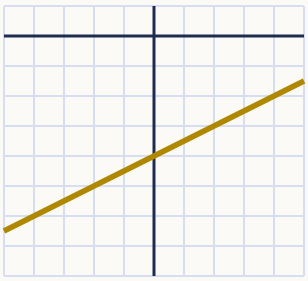


m =

answer: −2

6. *y-intercept*

In y = ½x − 4, what is the y-intercept?



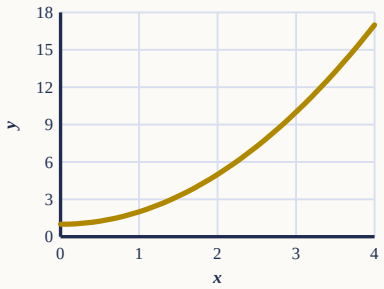
b =

answer: −4

7. *Linear?*

8. *Read a graph*

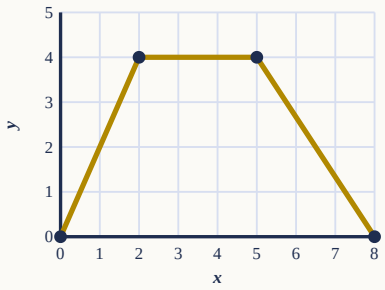
Is $y = x^2 + 1$ linear or nonlinear?



Answer:

answer: nonlinear

Look at the graph. From x = 2 to x = 5, is the graph increasing, decreasing or constant?



Answer:

answer: constant

SCORE 4.0 Target Content

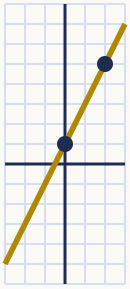
NC.8.F.3 · NC.8.F.4 · NC.8.F.5 Find slope and y-intercept from two points and from a table.

• *I can find the slope and y-intercept from two points or a table of values.*

Find each slope or y-intercept.

9. *Slope*

Find the slope through (0, 1) and (2, 5).

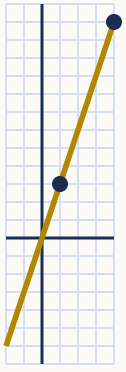


m =

answer: 2

10. *Slope*

Find the slope through (1, 3) and (4, 12).

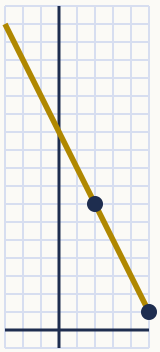


m =

answer: 3

11. *Slope*

Find the slope through (2, 7) and (5, 1).

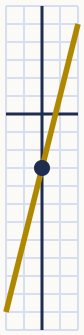


m = _____

answer: −2

12. *y-intercept*

A line has slope 4 and passes through (0, −3). Find the y-intercept.

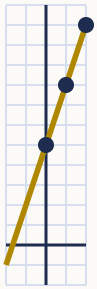


b = _____

answer: −3

13. *From a table*

x: 0, 1, 2 and y: 5, 8, 11. Find the slope.

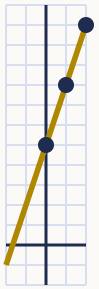


m = _____

answer: 3

14. *From a table*

x: 0, 1, 2 and y: 5, 8, 11. Find the y-intercept.



b = _____

answer: 5

15. *From a table*

Is this table linear or nonlinear?

x	0	1	2	3
y	1	2	5	10

Answer: _____

16. *From a table*

Is this table linear or nonlinear?

x	0	2	4	6
y	3	7	11	15

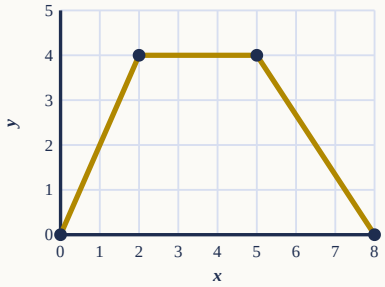
Answer: _____

answer: nonlinear

answer: linear

17. *Read a graph*

On which part is the graph decreasing?



Answer:

answer: x = 5 to x = 8

SCORE 5.0 Advanced Content

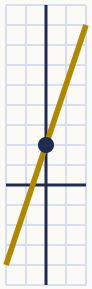
NC.8.F.2 · NC.8.F.3 · NC.8.F.5 · NC.8.EE.8 Write equations, compare functions, interpret slope, and solve a system by graphing.

• *I can write a linear equation, compare two functions, interpret slope as a rate of change, and read the solution of a system from its graph.*

Answer each question.

18. *Write equation*

A line has slope 3 and passes through (0, 2). Write y = mx + b.

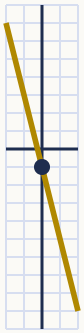


y =

answer: y = 3x + 2

19. *Write equation*

A line has slope −4 and passes through (0, −1). Write y = mx + b.

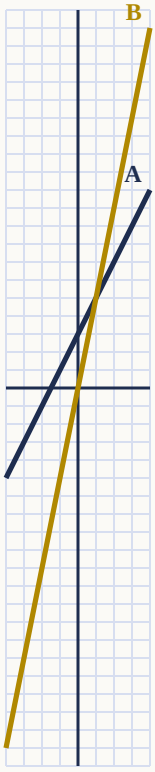


y =

answer: y = −4x − 1

20. *Compare*

Function A: y = 2x + 3. Function B: slope 5. Which has the greater slope? (A or B)

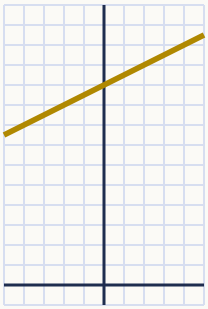


Answer:

answer: B

21. *Rate*

In y = 0.5x + 10, how much does y increase when x goes up by 1?



y increases by

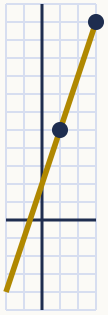
answer: 0.5

22. *Slope*

Find the slope through (1, 5) and (3, 11).

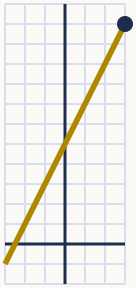
23. *y-intercept*

A line has slope 2 and passes through (3, 11). Find the y-intercept.



m =

answer: 3

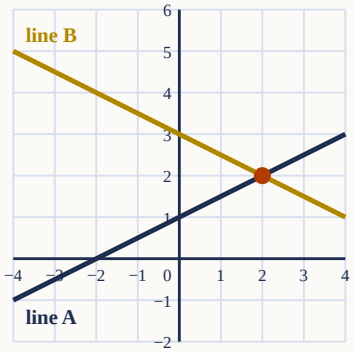


b =

answer: 5

24. *System*

Lines A and B are graphed. Write the solution of the system as an ordered pair.



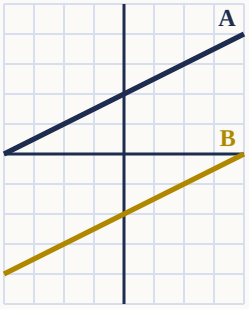
1 square = 1 unit

solution =

answer: (2, 2)

25. *System*

Lines A and B are graphed. How many solutions does this system have?



Answer:

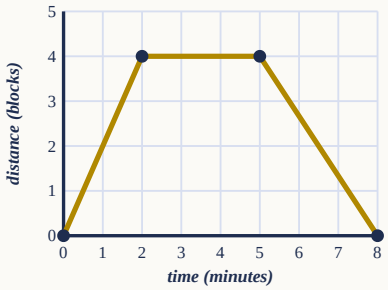
answer: none

26. *Graph story*

Mia walks to the store, shops, and walks home. The graph shows her distance from home. What does the flat part mean?

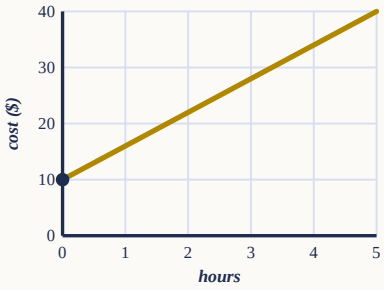
27. *Slope in a story*

A bike rental costs y = 6x + 10 dollars for x hours. What do the 6 and the 10 mean?



Answer:

answer: Mia is at the store. Her distance is not changing.



Answer:

answer: $6 for each hour, plus a $10 starting fee